From Data to Solutions · Weekend 0

scikit-learn

one workflow, not an algorithm zoo

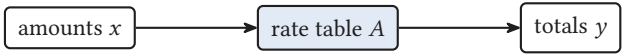
Carlos Cotrini

Section 1

From applying a rule to learning one

1

This morning you were handed the rule



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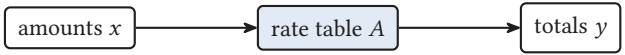

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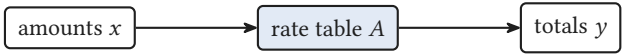

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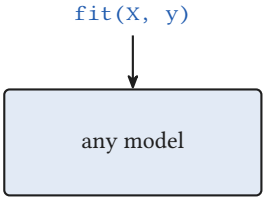
machine learning: learn the rule from examples

scikit-learn is one workflow



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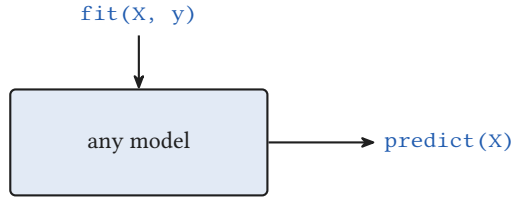

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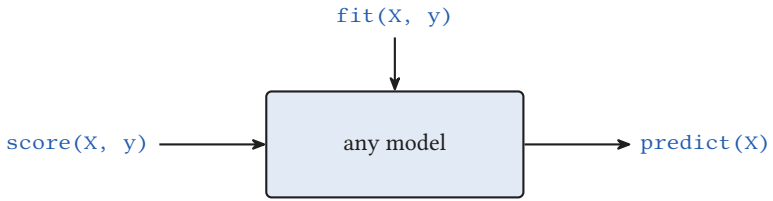

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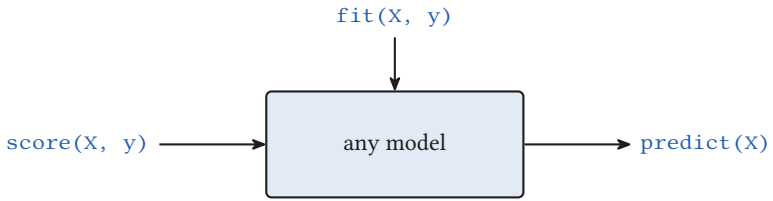

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learn the workflow, not the zoo

Section 2

The estimator API

2

You could hand-roll this fit

```
for step in range(2000):
  yhat = w*shots + b
  err = yhat - price
  w = w - lr*(err*shots).mean()
  b = b - lr*err.mean()
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gradient descent, by hand, with the NumPy you have

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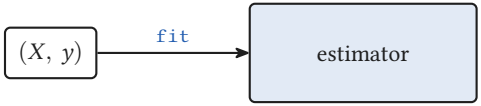
gradient descent, by hand, with the NumPy you have

same best-fit line, one call (sklearn picks the solver)

Three verbs, one object



`fit` learn from labelled examples

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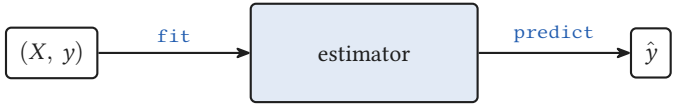

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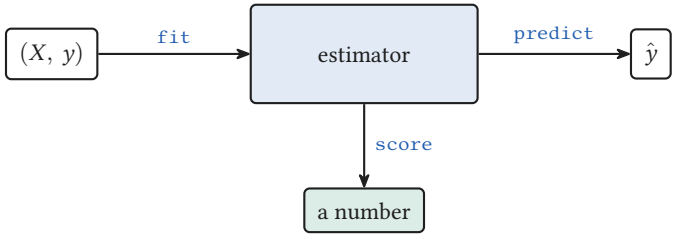

`fit` learn from labelled examples

`predict` apply to new inputs

`score` grade against the truth

The machine recovers your price rule

$$\text{price} = \underbrace{2.00}_{\text{base cup}}$$

The machine recovers your price rule

$$\text{price} = 2.00 + \underbrace{0.91}_{\text{per shot}} \text{ shots}$$

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true rule was $2.00 + 0.90\,\text{shots} + 0.004\,\text{milk}$

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- learned from **180** receipts

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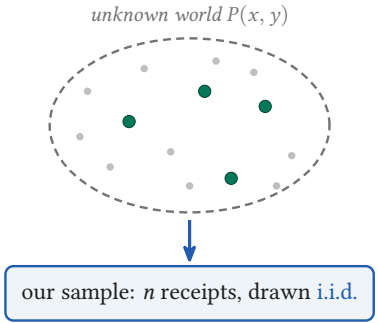
you stopped hand-writing the optimisation

A word from learning theory



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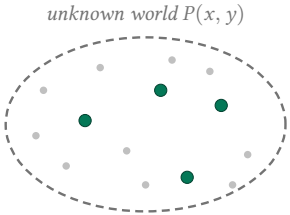

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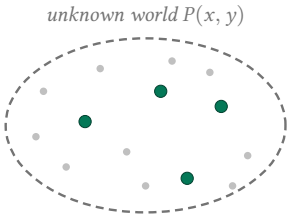

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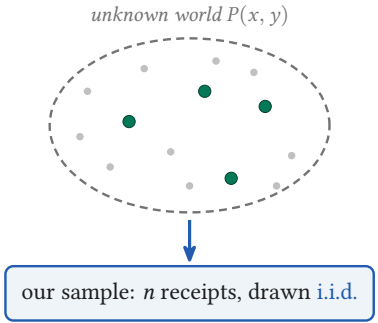

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search the candidate rules (ERM)

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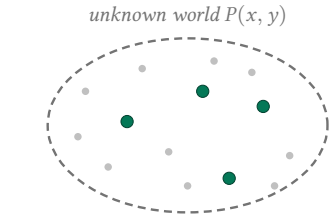

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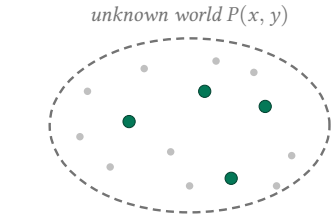

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`fit` minimises sample error; we really want small future error

Learning theory is the three verbs

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`fit` minimises sample error; `score` on test estimates the future

Section 3

The cardinal rule

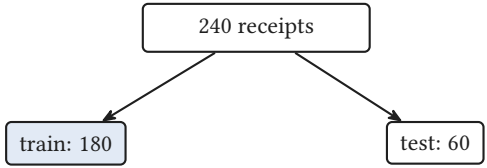
3

Never grade on what you trained on

240 receipts

- one call: `train_test_split(X, y)`

Never grade on what you trained on



- one call: `train_test_split(X, y)`
- R^2 on train $= 0.98$, on test $= 0.98$ here

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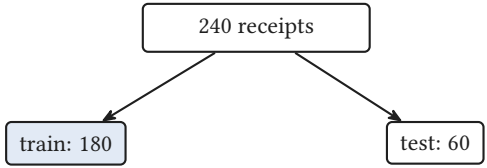

`fit` here

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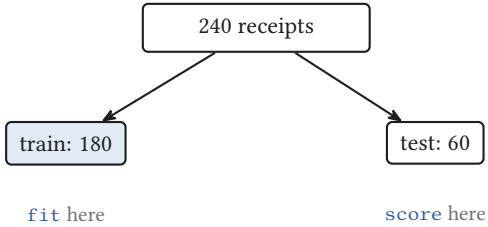

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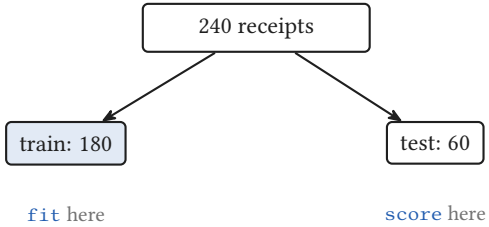

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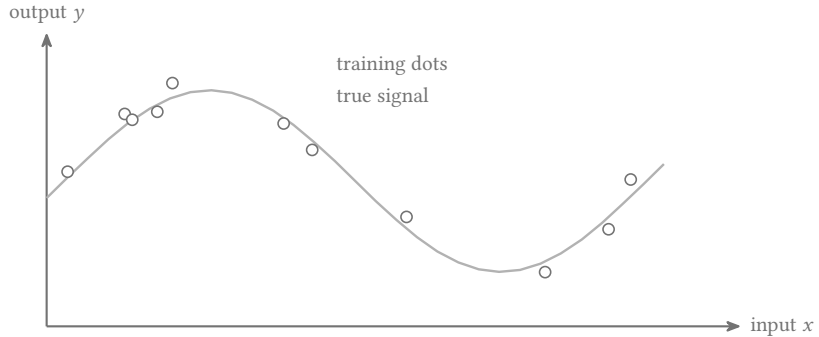
report the score on data the model never saw

Overfitting, drawn



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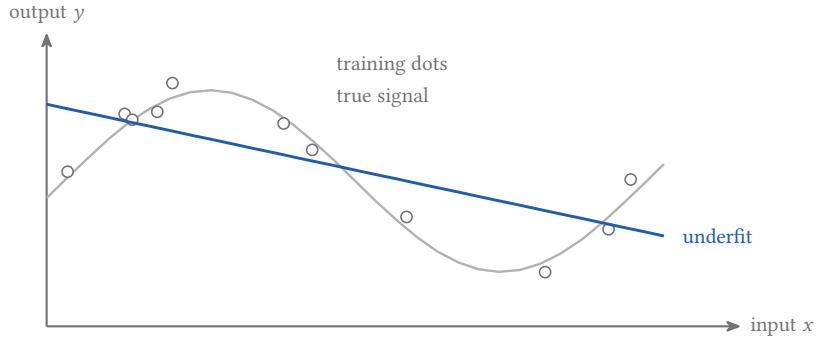

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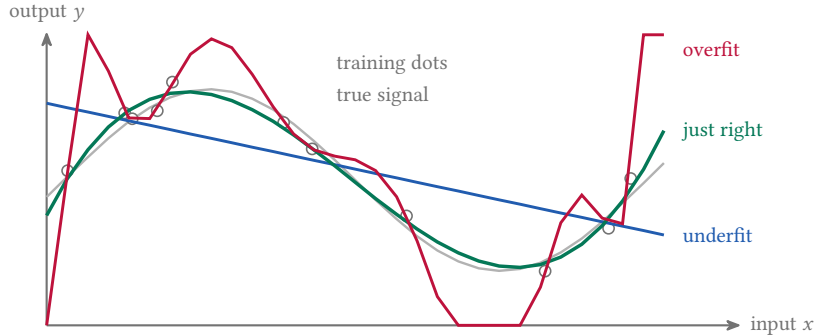

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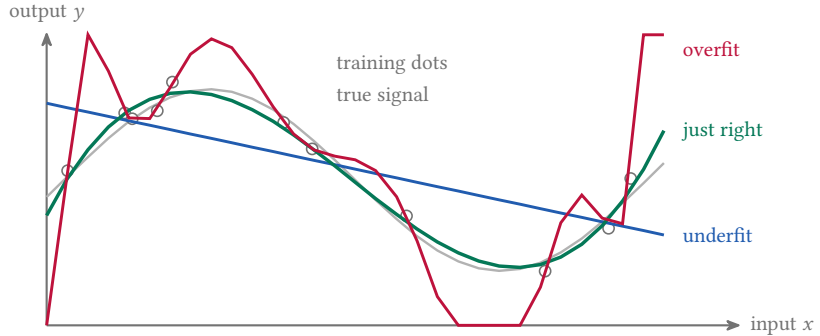
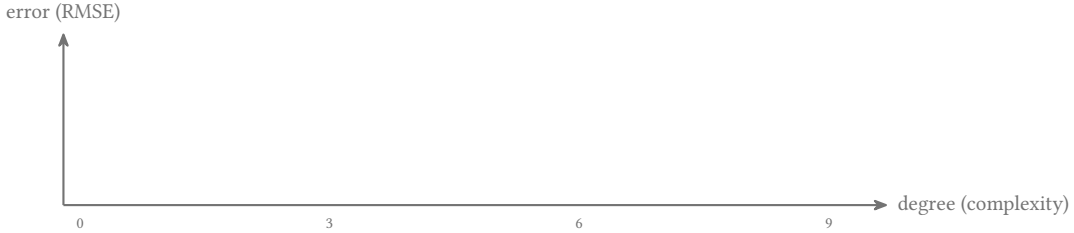


degree 9 aces the points it has seen, and fails on the next one

Why hold data out



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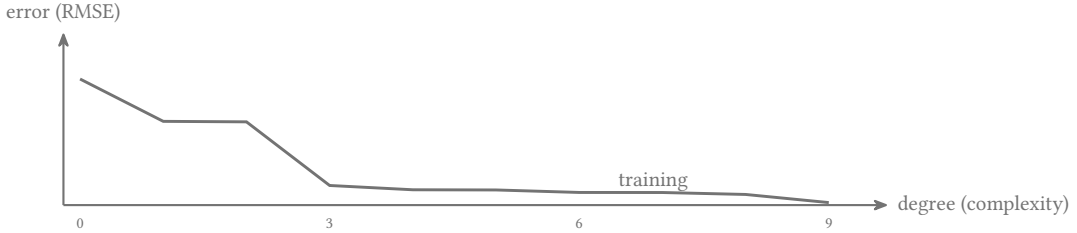

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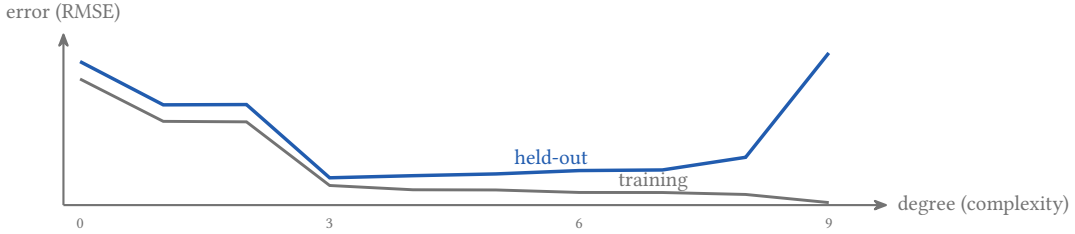

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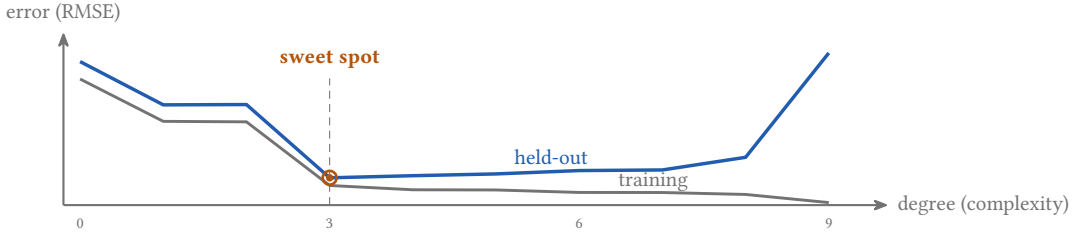

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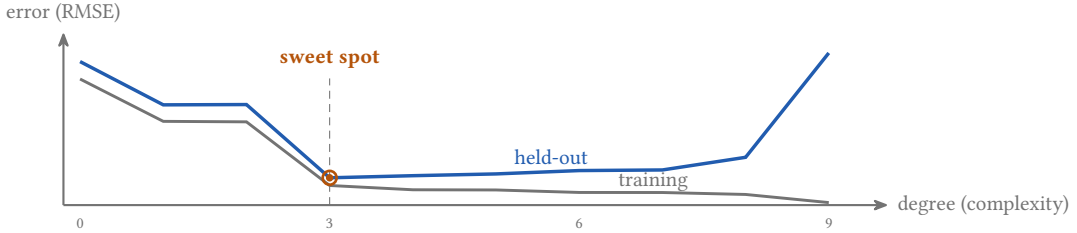

- training error always falls: blind to overfitting

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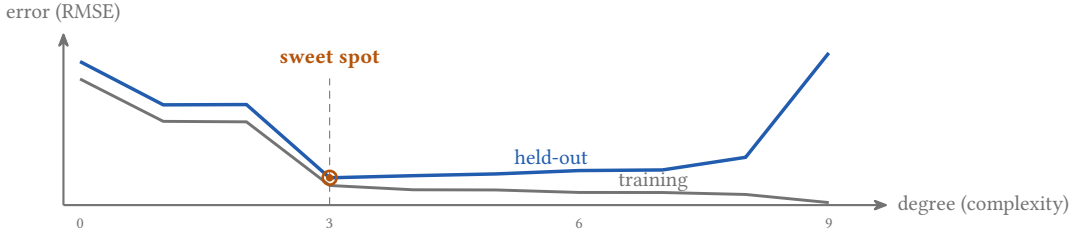

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- held-out error turns up: it can see it

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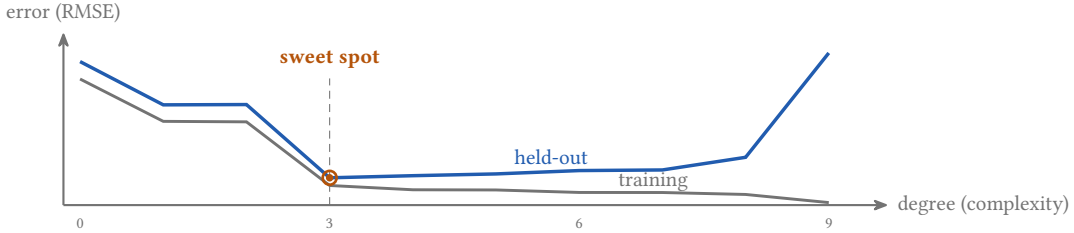

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- pick the degree at the U minimum (what `cross_val_score` estimates)

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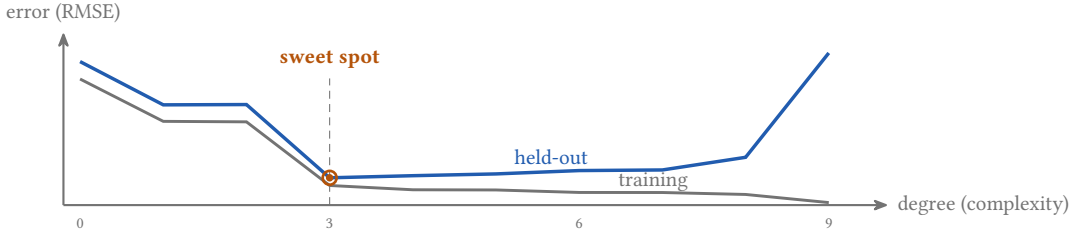

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training error always falls; only held-out error reveals overfitting

Section 4

A number becomes a yes/no

4

Same table, a new question

shots	milk	price	five_star
2	150	4.47	no
3	270	5.72	yes
1	0	2.88	no

- **regression**: predict a number (price)

A number becomes a yes/no

Same table, a new question

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- **regression**: predict a number (price)
- **classification**: predict a yes/no (a 5-star review)

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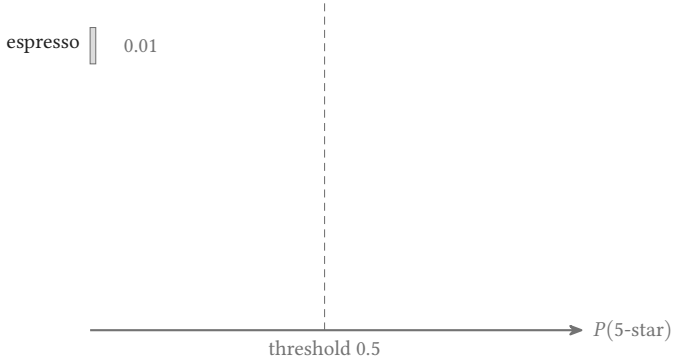
- **regression**: predict a number (price)
- **classification**: predict a yes/no (a 5-star review)
- `LogisticRegression().fit(X, y)`: same three verbs

A number becomes a yes/no

A probability, not just a label



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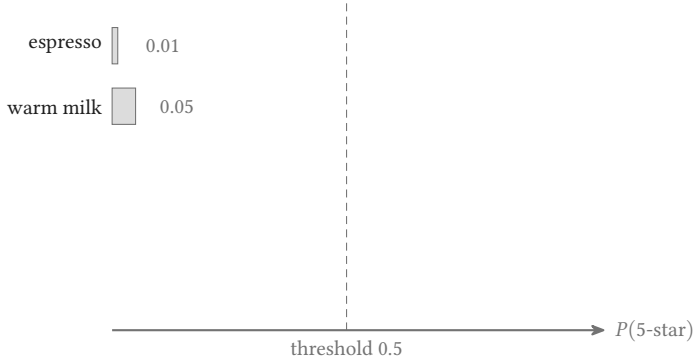

A number becomes a yes/no

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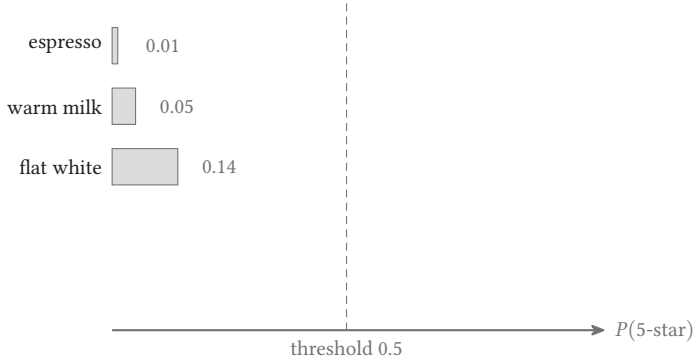
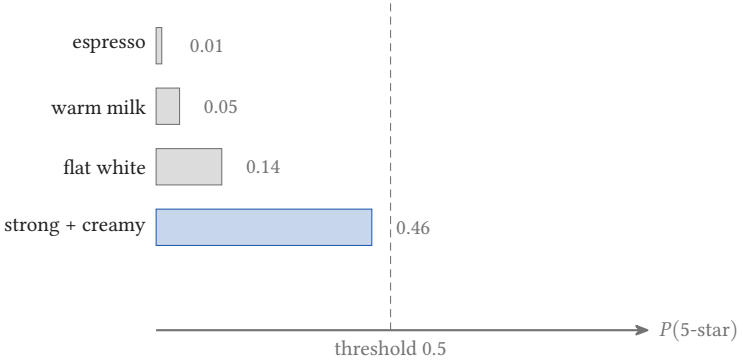


A probability, not just a label



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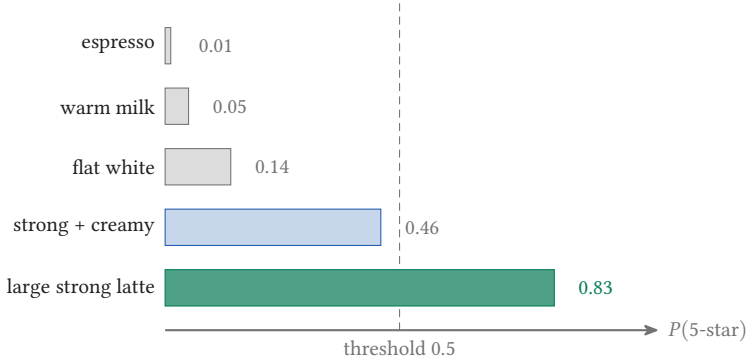
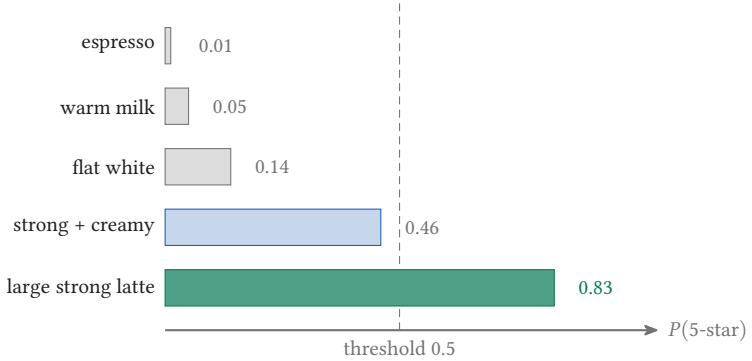
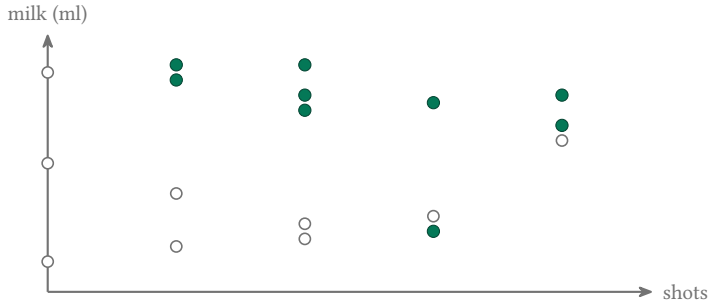


A probability, not just a label



`predict_proba` grades how sure, not just yes/no

The whole model in one picture



A number becomes a yes/no

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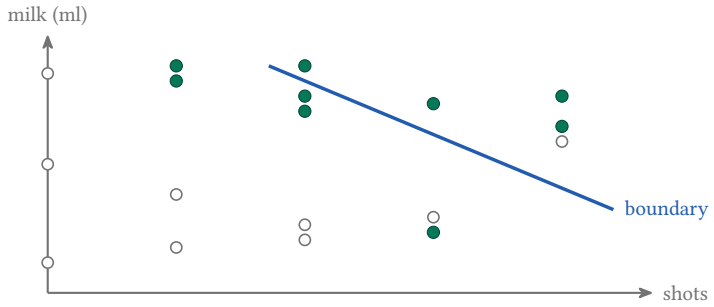
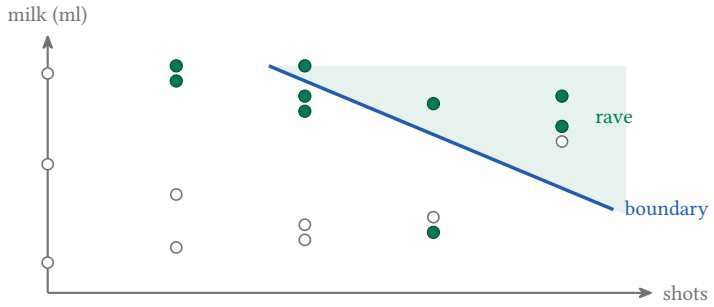


The whole model in one picture



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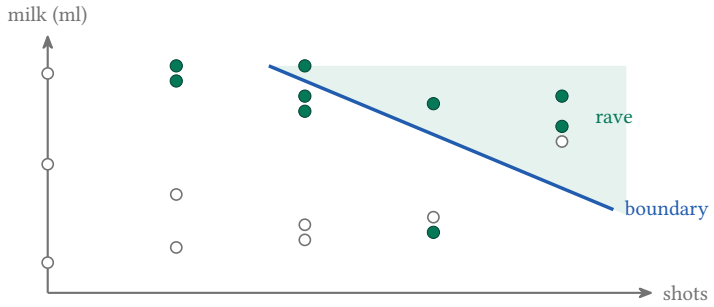

logistic regression: a line, squashed to a probability

Section 5

Accuracy can lie

5

Accuracy can lie

Most drinks do not get five stars

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"always predict NO" is 73% accurate

and catches zero of the ravers

Accuracy can lie

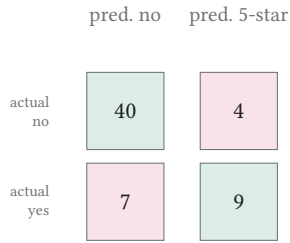
Most drinks do not get five stars

"always predict NO" is 73% accurate

and catches zero of the ravers

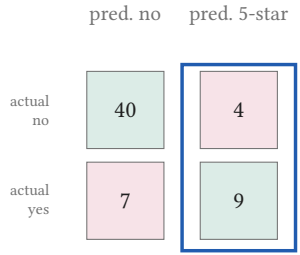
73% accurate and completely useless

Read the matrix, not the accuracy



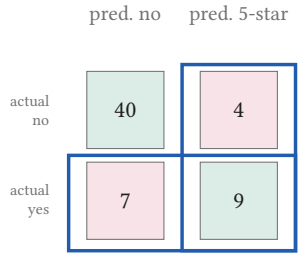
- accuracy 0.82, sounds fine

Read the matrix, not the accuracy



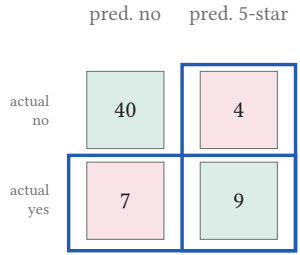
- accuracy 0.82, sounds fine
- precision 0.69: flagged raves that were real

Read the matrix, not the accuracy



- accuracy 0.82, sounds fine
- precision 0.69: flagged raves that were real
- recall 0.56: real raves it actually caught

Read the matrix, not the accuracy



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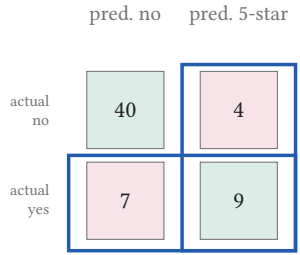
- ROC-AUC 0.85: one honest number

Read the matrix, not the accuracy



- accuracy 0.82, sounds fine
- precision 0.69: flagged raves that were real
- recall 0.56: real raves it actually caught
- ROC-AUC 0.85: one honest number

it misses nearly half the ravers: this is what the 2pm churn task grades

F1: one number when you need both

$$F_1 = \frac{2\,P\,R}{P + R}$$

F1: one number when you need both

$$F_1 = \frac{2 \times 0.69 \times 0.56}{0.69 + 0.56} = 0.62$$

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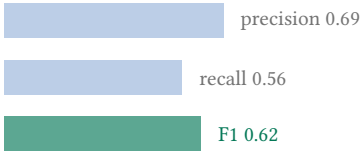
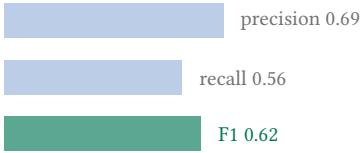


F1: one number when you need both

$$F_1 = \frac{2 \times 0.69 \times 0.56}{0.69 + 0.56} = 0.62$$



the harmonic mean: low if EITHER one is low

Section 6

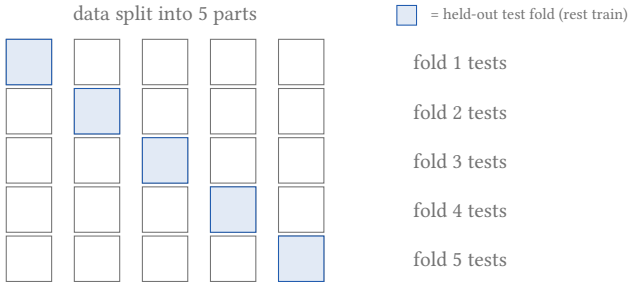
Leak-proof, and any model

6

Cross-validation: trust k splits, not one

- one split scores on just 60 drinks: noisy

Cross-validation: trust k splits, not one



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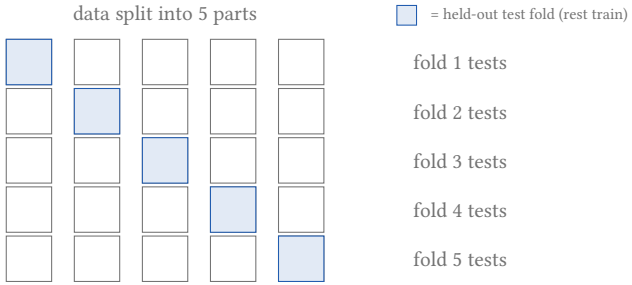
- 5 folds: each takes a turn as the test set, the rest train

Cross-validation: trust k splits, not one



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- `cross_val_score` averages the five scores

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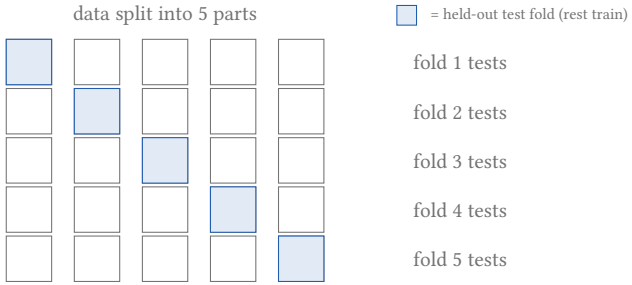


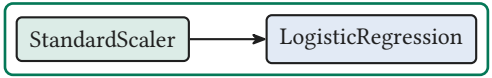
- one split scores on just 60 drinks: noisy
- 5 folds: each takes a turn as the test set, the rest train
- `cross_val_score` averages the five scores

5-fold ROC-AUC 0.85 ± 0.05, a stabler number than one split

Preprocess inside the pipeline

- scaling on the full data peeks at the test set

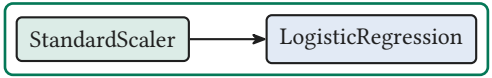
Preprocess inside the pipeline



one Pipeline, fit on the train fold only

- scaling on the full data peeks at the test set
- `make_pipeline(StandardScaler(), LogisticRegression())`

Preprocess inside the pipeline



one Pipeline, fit on the train fold only

- scaling on the full data peeks at the test set
- `make_pipeline(StandardScaler(), LogisticRegression())`
- `cross_val_score`: ROC-AUC 0.85 ± 0.05

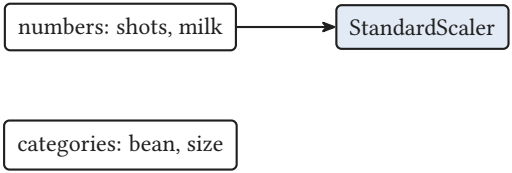
ColumnTransformer: each column type its own prep

numbers: shots, milk

categories: bean, size

- numbers want scaling; categories want one-hot

ColumnTransformer: each column type its own prep



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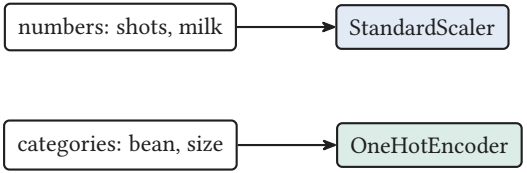

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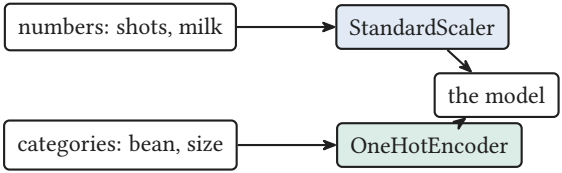

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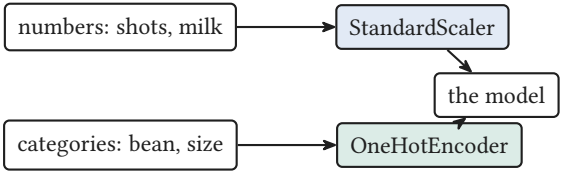

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one transform per column type, then fit as usual

Leak-proof, and any model

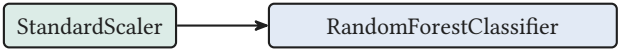
Swap one word, same workflow

`LogisticRegression()` AUC 0.85

Leak-proof, and any model

Swap one word, same workflow

`RandomForestClassifier()` AUC 0.78



Swap one word, same workflow

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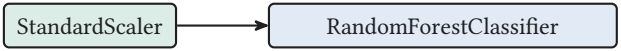

same `fit` · `predict` · `score` for every model (here the line wins, do not oversell the forest)

Section 7

The workflow you keep

7

The card you keep

- **split first**, before you look at anything

The card you keep

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- `fit` on train, `predict`, `score` on test

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scikit-learn is one workflow, and you now own it